

What It Takes To DESIGN And MANUFACTURE An INDIAN MICROPROCESSOR

There have been several acquisitions in the microprocessor and microcontroller industry recently. Let us take a look at the state of affairs and how the industry is facing. Designing, manufacturing, and selling a new processor is not easy, but open source seems to help



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We have x86 processors made by Intel and AMD, and ARM processors by various IC vendors who make them in multiple flavours. These are the only high-end microprocessors at present. Whereas 8- and 16-bit microcontrollers are available from multiple vendors like Microchip, Texas Instruments, NXP, ST, and Infineon, to name a few.

But during the last two years, the high-end microprocessors—especially with x86 architecture—have started facing competition from ARM based controllers in power consumption, speed, and multiple cores. This has put huge pressure on the legacy micropro-

cessors, which are the mainstay for PCs and servers.

However, Apple has broken their monopoly in PCs by switching over to ARM based controllers, which deliver far superior performance, putting pressure on the x86 processor family. But ARM is not a semiconductor as an integrated circuit but rather an IP (intellectual property) core, which the licensees can customise to create devices as per their needs.

This has led to proliferation of ARM based products across the board and major beneficiaries are mobile phones and embedded systems (not the PCs and servers!). Things had been going fine but suddenly the company ARM was bought over by the graphics processing units (GPUs) producer NVIDIA, leading to a major consternation. This has left every semiconductor company wondering what is going on.

The current situation

Since fast few years, x86 based processors are being challenged by the ARM based controllers in critical aspects like power, speed, size, and number of cores, with ARM based processors capturing the complete mobile market. Also, software for the ARM processors was never an issue as the volume of mobiles was much higher than of the PCs shipped.

Adding to this was ARM's business model of just licensing IP and leaving the realisation of devices to the vendors, which gave a lot of freedom to the mobile vendors. In addition, next-generation

